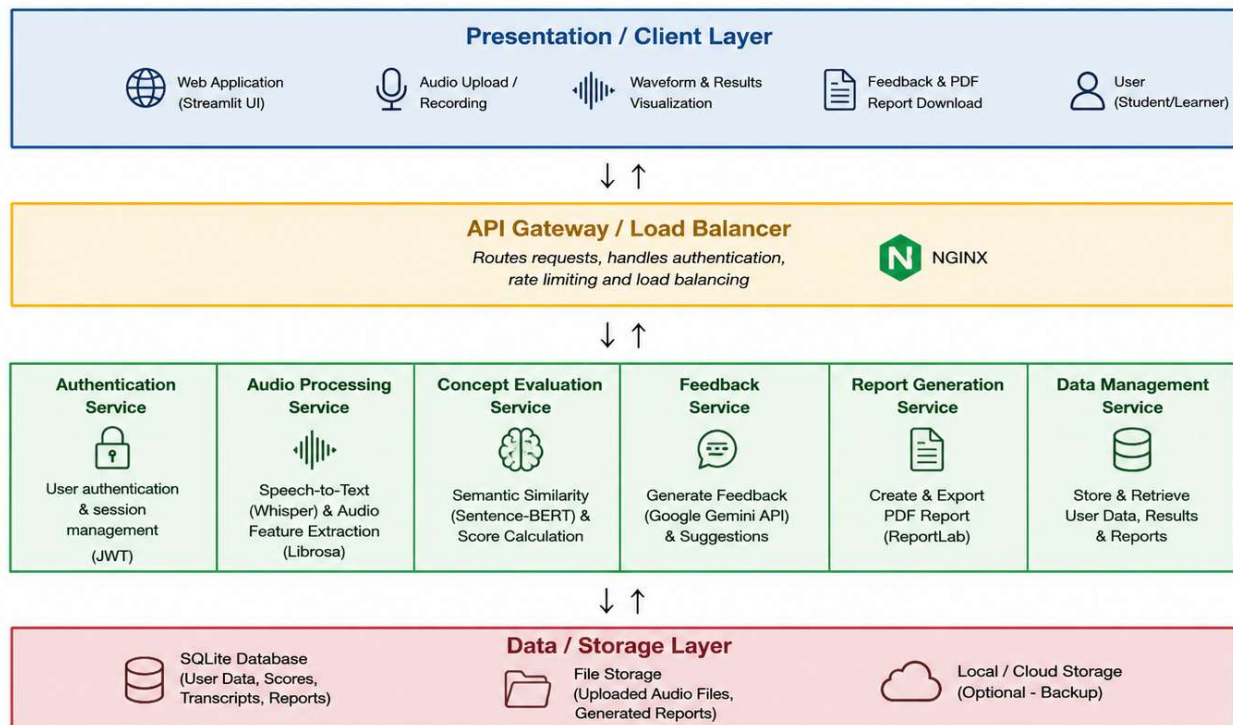


Solution Architecture

Date	06 July2026
Team ID	XXXXXX
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides user interface for audio upload/recording, visualization of results, feedback display and report download.	Streamlit, HTML/CSS, JavaScript
API Gateway / Load Balancer	Single entry point for all client requests. Routes requests to appropriate services and handles load balancing, rate limiting and security.	NGINX
Microservices (Application Layer)	Independent services that handle specific tasks such as authentication, audio processing, evaluation, feedback generation, report creation and data management.	Python, Flask (APIs), Whisper, Librosa, Sentence-BERT, Google Gemini API, ReportLab
Data / Storage Layer	Stores all persistent data including user information, transcripts, scores, reports and uploaded audio files.	SQLite, Local File Storage (Optional: AWS S3)

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for uploading audio, viewing waveform, analysis results, feedback, and downloading PDF reports.	Streamlit, HTML, CSS
API Gateway	Receives user requests and routes them to the appropriate backend services for speech processing and evaluation.	FastAPI
Microservices (Application Layer)	Performs speech-to-text transcription, semantic similarity analysis, audio feature extraction, score calculation, AI feedback generation, and PDF report generation.	Python, OpenAI Whisper, Sentence-BERT, Librosa, Google Gemini API, ReportLab
Database (Data Layer)	Stores uploaded audio information, transcripts, evaluation scores, feedback, reports, and session data.	SQLite